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MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Sections 1.1 and 1.2

Systems of Linear Equations
Row Reduction and Echelon Forms

What Is Linear Algebra?

Definition: (Merriam-Webster)
Linear Algebra is a branch of mathematics that is concerned with mathematical structures closed under the operations of addition and scalar multiplication and that includes the theory of systems of linear equations, matrices, determinants, vector spaces, and linear transformations.

Systems of Linear Equations

A linear equation is an equation which can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

A system of linear equations is a collection of one or more linear equations involving the same variables $x_1, x_2, \dots, x_n$.

A solution of the system is an ordered n -tuple, and the solution set of the system is the collection of all possible solutions of the system.

Systems of Linear Equations

A system of linear equations has no solutions, exactly one solution, or infinitely many solutions.

A consistent system has one or more solutions.

An inconsistent system has no solutions.

Example 1

Solve the system of linear equations using elimination.

$$x_1 + 2x_2 + 3x_3 = 4$$

$$4x_1 + 5x_2 + 6x_3 = 7$$

$$7x_1 + 8x_2 + 10x_3 = 10$$

Matrices

A matrix is a rectangular array of numbers.

The plural of matrix is matrices.

An $m \times n$ matrix has m rows and n columns.

Example 1 (continued)

Write the coefficient matrix and augmented matrix of the system.

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 4 \\4x_1 + 5x_2 + 6x_3 &= 7 \\7x_1 + 8x_2 + 10x_3 &= 10\end{aligned}$$

Elementary Row Operations

Replacement:

Replace one row by the sum of itself and a multiple of another row.

Interchange:

Interchange (swap) two rows.

Scaling:

Multiply all entries in a row by a nonzero constant.

Row Equivalence

Two matrices are called row equivalent if there exists a sequence of elementary row operations which transform one matrix into another.

If the augmented matrices of two linear systems are row equivalent, then they have the same solution set.

All elementary row operations are reversible!

Example 1 (continued)

Apply elementary row operations to the augmented matrix of the system to solve the system of linear equations.

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 4 \\4x_1 + 5x_2 + 6x_3 &= 7 \\7x_1 + 8x_2 + 10x_3 &= 10\end{aligned}$$

Echelon Forms

A rectangular matrix is in (row) echelon form if it has the following three properties:

1. All nonzero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it. (The leading entry of a row is the first nonzero entry.)
3. All entries in a column below a leading entry are zeros.

Echelon Forms

If a matrix in (row) echelon form also satisfies the following two conditions, then it is in reduced (row) echelon form:

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

Echelon Forms

Each matrix is row equivalent to one and only one reduced echelon matrix.

Echelon form is not unique, but any echelon form of a matrix will lead to the same reduced echelon form.

Pivot Positions

A pivot position in a matrix is a location in the matrix that corresponds to a leading 1 in the reduced echelon form of the matrix.

A pivot column of the matrix is a column which corresponds to a pivot position.

Even though echelon form is not unique, any echelon form of a matrix will result in leading entries in the same positions – the pivot positions.

The Row Reduction Algorithm

The Row Reduction Algorithm provides an efficient procedure for transforming a matrix into either echelon or reduced echelon form.

Example 2

Apply elementary row operations to transform the following matrix into reduced echelon form.

$$\begin{bmatrix} 0 & 2 & 4 & 6 & 8 \\ 1 & 3 & 5 & 7 & 9 \\ 3 & 6 & 9 & 12 & 14 \end{bmatrix}$$

The Row Reduction Algorithm

1. Begin with the leftmost nonzero column. This is a pivot column, and the pivot position is at the top.
2. Select your favorite nonzero entry in the pivot column as a pivot and, if necessary, interchange rows to place this entry in the pivot position.
3. Use row replacement operations to create zeros in all positions below the pivot.

The Row Reduction Algorithm

4. Ignore the row containing the pivot and all rows (if any) above it. Then, repeat steps 1-3 for the remaining submatrix. Continue this process until there are no more nonzero rows to modify.

At this point, the matrix is in echelon form. One more step will place the matrix in reduced echelon form.

The Row Reduction Algorithm

5. Beginning with the rightmost pivot and working to the left, apply scaling operations to make each pivot a 1 and use replacement operations to create zeros above each pivot.

The matrix is now in reduced echelon form.

Example 2 (continued)

Apply elementary row operations to transform the following matrix into reduced echelon form.

$$\begin{bmatrix} 0 & 2 & 4 & 6 & 8 \\ 1 & 3 & 5 & 7 & 9 \\ 3 & 6 & 9 & 12 & 14 \end{bmatrix}$$

Now, assume this matrix is the augmented matrix of a system of linear equations. Describe the solution(s), if any.

Basic and Free Variables

The variables corresponding to pivot columns in an augmented matrix are called basic variables.

The remaining variables, if any, are called free variables.

Example 3

Find the general solution of the linear system whose augmented matrix has been row reduced to

$$\begin{bmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & -3 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Existence and Uniqueness

We wish to address two fundamental questions about a linear system:

1. Does at least one solution exist?
2. If a solution exists, is it unique?

Existence and Uniqueness Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is not a pivot column.

If the linear system is consistent, then it will have exactly one solution if there are no free variables.

If there are one or more free variables in a consistent system, it will have infinitely many solutions.
